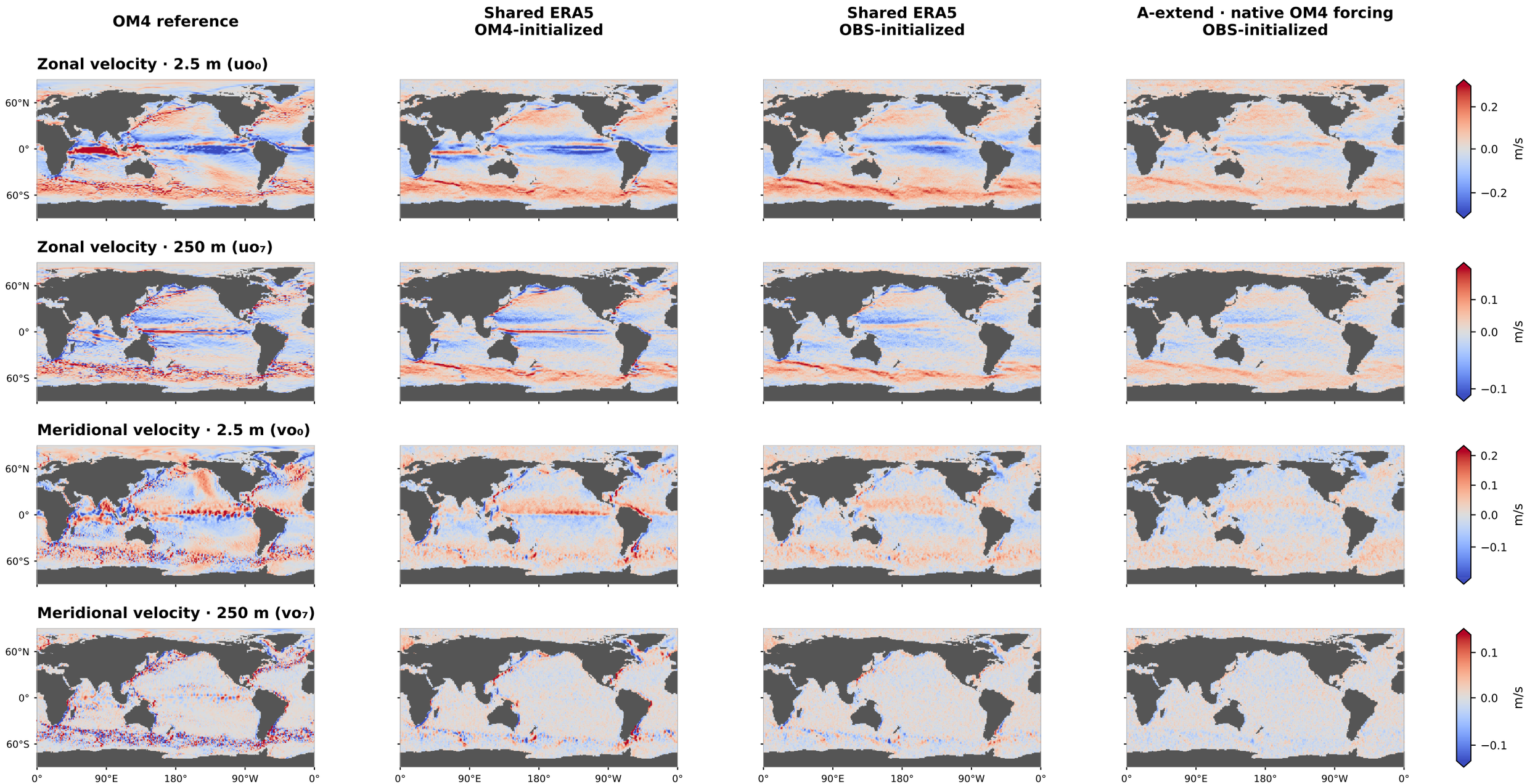
Ocean velocity | Initialization · before cooldown · 8-member mean

Date-matched five-day fields: 27 Dec 2014–1 Jan 2015 (exclusive end) · EMA weights · eight-member diffusion mean



Both models: 4k mixed updates, before OBS cooldown. Shared ERA5 versus native OM4 forcing during model-data training.

“Native OM4 forcing” describes A-extend’s OM4 training tasks. All OBS-initialized examples use ERA5 through their learned adapters.

OBS velocity outputs are full-77 diagnostic decodes, not directly supervised OBS currents. OM4 reference is simulation context, not observational truth.

Each row uses OM4 reference 2nd–98th percentile bounds for all columns; tails clipped. Dark grey = dry cells. Eight decoder draws per fixed latent state; 32 Heun steps each.